PCBC H Parametrics Practice Test D Name _____ _____/12

Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: the calculator question comes first, as announced in class. Every question copies a homework problem with new numbers. 3 points each is a practice split of the posted 12.

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth. *(practice instructions)*

1. (a) Graph the cycloid defined by $x = 4t - 4\sin t,\ y = 4 - 4\cos t$ for t in $[0, 4\pi]$. Sketch it and write the window you used.

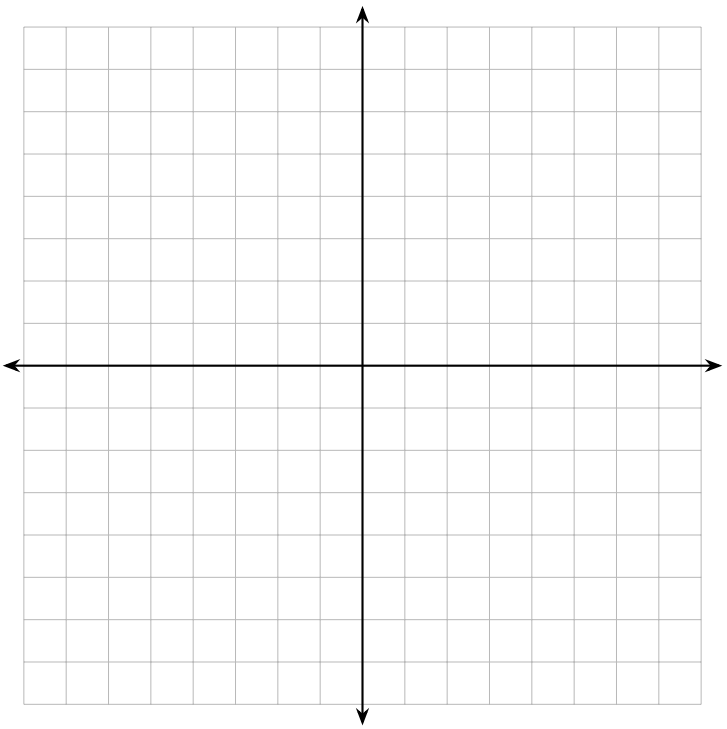
(b) The graph of $x = 3\cos t,\ y = 2\sin 2t$ is a Lissajous figure. Graph the figure for t in $[0, 6.5]$ in the window $[-4, 4]$ by $[-3, 3]$ and sketch it. **3 pts**

Non-Calculator 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

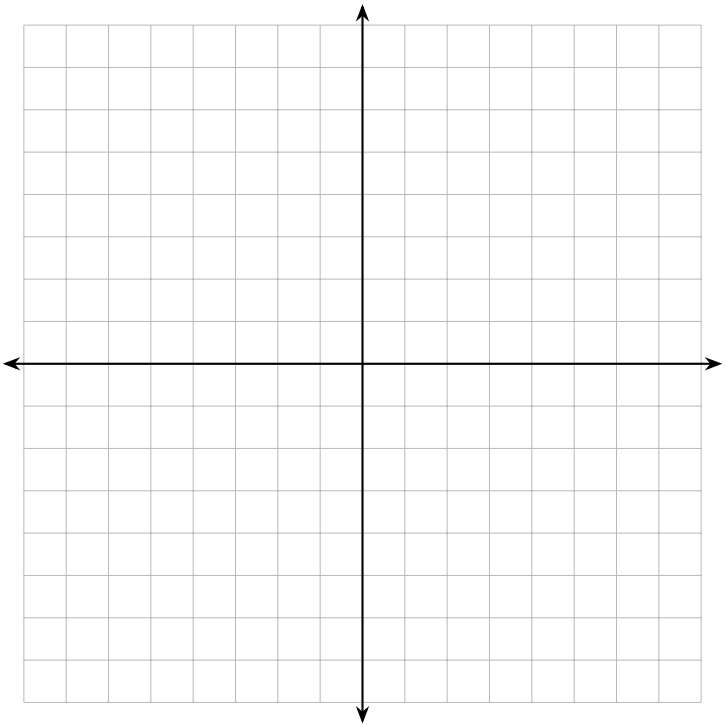
Find a rectangular equation for the plane curve, and graph the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

2. $x = t^2 + 2, \quad y = t^2 - 5, \quad$ for t in $(-\infty, \infty)$ **3 pts**



Find a rectangular equation for the plane curve, and graph the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

3. $x = 2t, \quad y = \sqrt{t^2+4}, \quad$ for t in $(-\infty, \infty)$ **3 pts**



Consider the parametric equations $x = f(t)$, $y = g(t)$ for t in $[a,b]$, with $c > 0$. Answer each part. (3 points)

4. How is the graph affected if: **3 pts**

(a) $x = f(t)$ is replaced by $x = c + f(t)$.

(b) $y = g(t)$ is replaced by $y = g(t) - c$.

(c) $x = f(t)$ is replaced by $x = f(t) - c$.

Choices for each: shifted c units left, right, up, or down.

PCBC H Parametrics Practice Test D: Answer Key

12 points

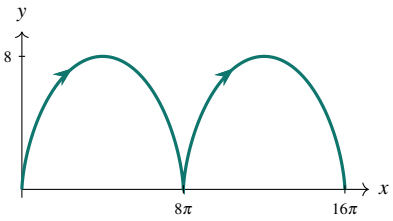
Each solution follows the method used in class. The gray box shows another way that also works. Once t is gone, any equivalent form of the answer is fine. Every value was checked in Python.

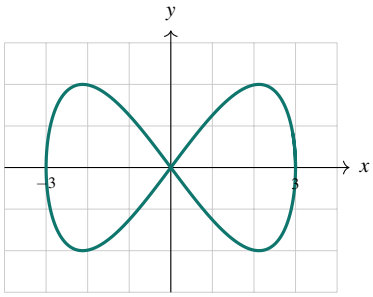
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1. *Calculator* (like homework day 2 #17 and #18)

(a) Two arches, each $8\pi \approx 25.133$ wide and 8 tall, touching the x-axis at 0, 8π, 16π. Window: x from 0 to 52, y from -1 to 9.
(b) A sideways figure eight through the origin, from $x = -3$ to 3 and $y = -2$ to 2.

In class	Why
Radian mode	t is inside sin and cos with no degrees.
Graphs, Parametric: $x1(t) = 4t - 4\sin(t)$, $y1(t) = 4 - 4\cos(t)$	Enter it exactly as written.
t from 0 to 4π	The interval given. Two arches.
Window x: 0 to 52, y: -1 to 9	Wide enough for $4(4\pi) \approx 50.3$; tall enough for 8.
(b) $x1(t) = 3\cos(t)$, $y1(t) = 2\sin(2t)$, t: 0 to 6.5	Use the window they give: $[-4,4]$ by $[-3,3]$.
Sketch what the screen shows	Copy the shape and where it crosses the axes.





Other way. Find the numbers without graphing: the arches meet the x-axis when $\cos t = 1$ ($t = 0, 2\pi, 4\pi$, so $x = 0, 8\pi, 16\pi$); the top is when $\cos t = -1$, $y = 4 + 4 = 8$.

Practice scoring (3 pts): 1.5 for the cycloid with a window that shows both arches, 1.5 for the Lissajous figure.

Trap: degree mode (the graph comes out flat and wrong), or a window too small to show the whole curve.

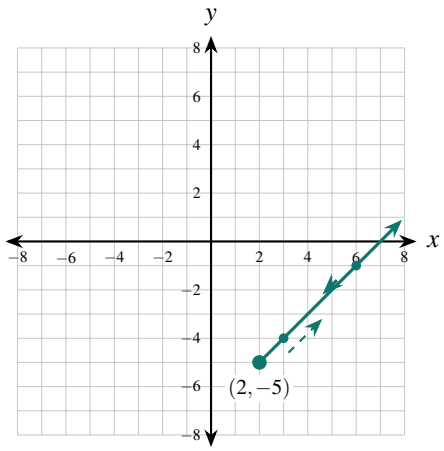
2. $x = t^2 + 2, \quad y = t^2 - 5, \quad$ for t in $(-\infty, \infty)$ *Non-Calculator* (like homework day 1 #5)

$\boxed{y = x - 7, \text{ for } x \text{ in } [2, \infty)}$
A ray starting at $(2, -5)$, solid dot there, arrowhead at the other end. It comes down to $(2, -5)$ and goes back up the same path.

In class	**Why**
t-chart: $t = -2, -1, 0, 1, 2$	Pick values on both sides of 0.
Plot: same points twice	t and $-t$ give the same point: it retraces.
$t^2 = x - 2$	Get t^2 alone instead of t (no $\pm$ needed).
$y = (x-2) - 5 = x - 7$	Substitute t^2 into y.
for x in $[2, \infty)$	Smallest x in the chart is 2; $t^2 \geq 0$, so x never goes below 2.

t	(x, y)
-2	$(6, -1)$
-1	$(3, -4)$
0	$(2, -5)$
1	$(3, -4)$
2	$(6, -1)$

Arrows: down toward $(2, -5)$, then back up. Only half of the line $y = x - 7$.



Other way. Subtract the equations: $x - y = (t^2 + 2) - (t^2 - 5) = 7$, so $y = x - 7$ in one line. The homework answer sheet draws the whole line; the t-chart shows x never goes below 2, so restrict it.

Practice scoring (3 pts): 1 for $y = x - 7$, 1 for x in $[2, \infty)$, 1 for the ray with a solid dot at $(2, -5)$ and arrows.

Trap: drawing the whole line. $x = t^2 + 2$ can't be less than 2.

3. $x = 2t, \quad y = \sqrt{t^2+4}, \quad$ for t in $(-\infty, \infty)$ *Non-Calculator* (like homework day 2 #9)

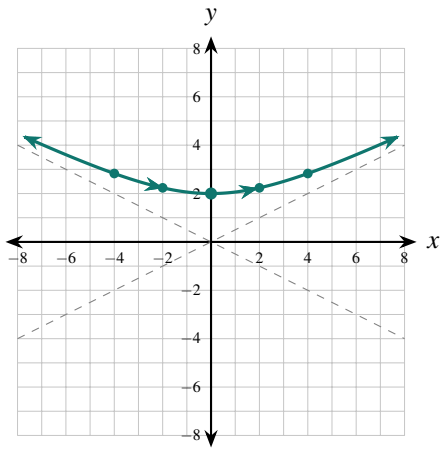
$\boxed{y = \sqrt{\frac{x^2}{4}+4}}$

x in $(-\infty, \infty)$. A hyperbola-shaped curve, lowest point $(0,2)$, traced left to right, arrowheads on both ends.

In class	Why
t-chart: $t = -2, -1, 0, 1, 2$	Parentheses with ∞: pick any t.
Plot, connect, arrows left to right	Lowest t to highest t.
$t = \frac{x}{2}$	Get t alone from x: one answer.
$y = \sqrt{\left(\frac{x}{2}\right)^2+4} = \sqrt{\frac{x^2}{4}+4}$	Substitute into y.
x in $(-\infty, \infty)$	$x = 2t$ can be any number, so nothing to restrict.

t	(x, y)
-2	$(-4, 2\sqrt{2})$
-1	$(-2, \sqrt{5})$
0	$(0, 2)$
1	$(2, \sqrt{5})$
2	$(4, 2\sqrt{2})$

$\sqrt{5} \approx 2.24$, $2\sqrt{2} \approx 2.83$. A square root is never negative: top branch only.



Other way. Square both sides to see the shape: $y^2 = \frac{x^2}{4} + 4$, so $\frac{y^2}{4} - \frac{x^2}{16} = 1$, the top branch of a hyperbola with asymptotes $y = \pm\frac{x}{2}$. Keep the $\sqrt{\ }$ form as the answer.

Practice scoring (3 pts): 1 for $y = \sqrt{x^2/4+4}$, 1 for the top branch only, 1 for the graph with left-to-right arrows.

Trap: writing $y = \pm\sqrt{\cdot}$ or drawing a bottom branch. The original y is a square root.

4. How is the graph affected if: *Non-Calculator* (like homework day 2 #22)

(a) shifted c units right (b) shifted c units down (c) shifted c units left

In class	Why
(a) $x = c + f(t)$: every x gets c bigger	Same y, bigger x: right.
(b) $y = g(t) - c$: every y gets c smaller	Same x, smaller y: down.
(c) $x = f(t) - c$: every x gets c smaller	Same y, smaller x: left.

Other way. Test it with a point: if $(f(t), g(t)) = (1,1)$ and $c = 2$, then (a) gives $(3,1)$, which is to the right.

Practice scoring (3 pts): 1 point each.

Trap: mixing this up with function shifts like $f(x+c)$, which go left. Here the c is added to the coordinate itself, so it moves the way it looks.